S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

RiskLens

Financial Event Risk Intelligence & Portfolio Stress Testing



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PROBLEM STATEMENT & VISION

The Problem & Our Objective

THE CHALLENGE

Unstructured Financial Information

Important risk signals can be buried in financial news and user-provided text.

Inconsistent Risk Assessment

Manually interpreting sentiment, event type, and potential severity can make risk assessment difficult to standardize.

Limited Portfolio-Level Context

Identifying a financial event is only part of the task; users also need to explore how portfolio holdings might respond under different scenarios.

OUR OBJECTIVE

"Transform financial text into structured risk signals and connect event analysis with portfolio stress testing."



TRANSFORMATION PATH: From financial information → to structured risk signals → to scenario-based portfolio analysis.

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TAKEAWAY: Financial Information → Structured Risk Signals → Scenario-Based Portfolio Analysis

TECHNICAL FOUNDATION

System Architecture & Technology Stack

1. INPUT SOURCES

Financial News
Articles & press releases

User Provided Text
Social media & custom prompts

2. INGESTION & QUALITY

Data Ingestion & Validation

Text Preprocessing
Normalization & Noise Removal

Data Quality Checks

3. RISK INTELLIGENCE ENGINE

- Entity Recognition ✓
- Sentiment Analysis ✓
- Event Classification ✓
- Impact Scoring & Risk Level ✓

OUTPUT ARTIFACT
Structured Risk Signal

4. STRESS TESTING

Portfolio Management API

Stress Testing Engine
Predefined & Custom Shocks

Estimated Scenario Impact

5. APPLICATION & API

React Dashboard

FastAPI REST Backend

MongoDB Store

CLIENT READY

TECH STACK: Frontend: React 18 + Vite | Backend: Python + FastAPI | Database: MongoDB | Formats: JSON / CSV / YAML

Implementation & Engineering Challenges

SECTION A — IMPLEMENTATION HIGHLIGHTS

Financial Text Processing

- Input validation & automated text preprocessing.
- Analysis of entities, sentiment & financial event categories.
- Generation of structured risk signals with impact & confidence scores.

API-Driven Integration

- React/Vite frontend connected to Python/FastAPI backend.
- REST endpoints supporting ingestion, analysis & stress testing.
- MongoDB storage for supported application persistence.

Portfolio Scenario Analysis

- Holdings represented via structured asset information.
- Predefined stress scenarios & custom user shocks.
- Scenario-based estimates to evaluate potential exposure.

SECTION B — CHALLENGES & RESPONSES

CHALLENGE 1 — **Unstructured Input**

Financial text may vary significantly in format and content quality.

Response: Apply input validation, preprocessing, and data-quality checks.

CHALLENGE 2 — **End-to-End Integration**

Risk analysis, portfolio operations, and stress testing need to work seamlessly.

Response: Organize functionality through modular backend APIs and a unified dashboard.

CHALLENGE 3 — **Interpreting Scenario Results**

Stress-test results depend on selected assumptions and portfolio inputs.

Response: Present results as scenario-based estimates rather than actual market predictions.

ENGINEERING PRINCIPLE: Prioritize structured API communication, transparent scenario assumptions, and rigorous input handling over uncalibrated forecasts.

PROTOTYPE DEMONSTRATION

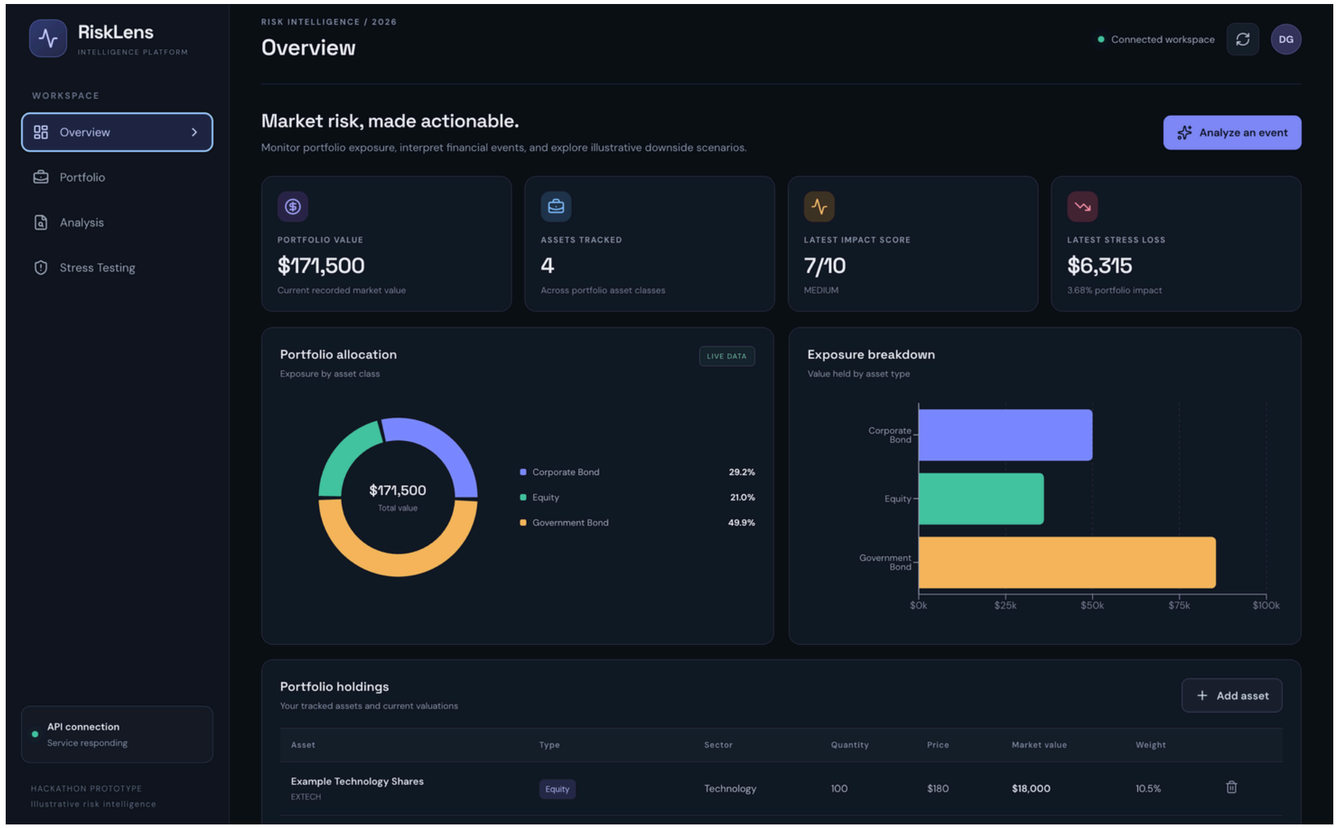
Results & Working Prototype

1. FINANCIAL EVENT ANALYSIS

LIVE UI OUTPUT

SUBMITTED FINANCIAL TEXT INPUT:

"Federal regulators initiate inquiry into tech supply chain delays; key semiconductor component imports down 18% Q/Q."



LINK : [Click here]

Impact, Future Scope & Conclusion

PROJECT IMPACT

- Connects financial-event analysis with portfolio stress testing in one workflow.
- Converts financial text into structured risk signals for easier examination.
- Enables users to explore potential portfolio exposure under selected stress scenarios.

LIMITATIONS & FUTURE SCOPE

- Current: Analysis depends on implemented classification logic and available input data.
  - └ Future: Evaluate and improve the analysis using a larger, labelled financial-event dataset.
- Current: Stress-test results depend on portfolio information and scenario assumptions.
  - └ Future: Expand scenario coverage and validate assumptions against suitable historical data.
- Current: RiskLens is a prototype, not a production-grade financial decision system.
  - └ Future: Explore stronger monitoring, evaluation, and integration capabilities.

“Turning financial events into structured risk insights and portfolio-level scenario analysis.”

Thank You

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